

Khaled Bin Walid

☎ +1-315-603-3403 ✉ walidk@clarkson.edu

in [khaled-823-d](#) 📖 [Khaled Bin Walid](#)

🏠 Potsdam, New York 13676, USA

Summary

Ph.D. candidate in Electrical and Computer Engineering with three years of research experience in electricity market design, power system reliability, and renewable energy integration. Skilled in optimization modeling, long-duration energy storage scheduling, and applying machine learning to demand and wind power forecasting. Brings 5+ years of industry experience as Senior Supply Chain Manager at Walton Hi-Tech Industries, leading cross-functional teams, developing forecasting models, and implementing strategies that optimize costs and enhance operational efficiency. Combines strong analytical and leadership skills with a multidisciplinary background bridging research and real-world applications in the evolving energy landscape.

Areas of Interest

Energy Market Design & Operation; Optimization Modeling; Energy Storage Scheduling & Renewable Energy Integration

Education

- **Clarkson University** **Potsdam, USA**
 - *Ph.D. Candidate in Electrical & Computer Engineering* *2023 – Present*
 - *M.Sc. in Electrical & Computer Engineering; CGPA: 3.97/4.00* *2023 – 2025*
- **University of Dhaka** **Dhaka, Bangladesh**
 - *M.Sc. in Electrical & Electronics Engineering; CGPA: 3.09/4.00* *2016 – 2018*
 - *B.Sc. in Electrical & Electronics Engineering; CGPA: 3.13/4.00* *2012 – 2015*

Technical Skills

- **Optimization & Power Systems:** Optimization modeling (Gurobi), Electricity Market Design & Operation, Energy Storage Scheduling
- **Machine Learning & Forecasting:** Data Analysis, Electricity Demand Forecasting, Wind Power Forecasting
- **Programming Languages:** C, C++, Python, Julia, MATLAB
- **Simulation Tools:** PLEXOS, Simulink, PSCAD, MultiSim, Proteus, CISCO Packet Tracer
- **Other Tools:** L^AT_EX, Microsoft Office, Project Management

Research Experience

- **Collaborative & Industry Projects**
 - **Value of Lost Load Study for New York Customers:** Sponsored by NYISO, 2025. Contributed to customization of VOLL estimates for individual utilities, extending beyond existing state-level estimations.
 - **Operational Energy Adequacy:** Sponsored by NYISO, 2025. Contributed to the design of an operational resource adequacy tool and a risk-informed CACT tool; validated results by building a synthetic New York grid model.
 - **Assessing the Social Benefits of Grid Capacity Expansion:** Sponsored by Avangrid, 2025. Collaborative project evaluating the social benefits of grid expansion in Avangrid service territory; contributed by assessing the value of lost load for NYSEG, UI, CMP, and RG&E.
 - **AIRU-WRF – AI-Powered Physics-based Tool for Offshore Wind Forecasting and Grid Integration:** Sponsored by NOWRDC/DOE (Project #133/192900), 2023–2025. Collaborative project with Rutgers University; contributed to grid integration studies and performed economic and reliability analysis.
 - **REU Site: Summer Research Experience on Resilient Carbon-Free U.S. Electric Power Systems:** Sponsored by NSF, 2022–2025. Guided REU summer students in 2024 and 2025 on research subtasks including uncertainty reserve design, ramping need analysis, resource adequacy modeling, production cost simulation, optimization and value of lost load assessment.
- **Academic & Independent Projects**
 - **Long Duration Energy Storage Arbitrage Modeling with Price Forecasting Uncertainty:** Developed a penalty-based SOC deviation mechanism integrated into SCUC to better align ISO dispatch with asset owner incentives under multi-day price uncertainty. Demonstrated improved storage utilization and economic efficiency in a New York power grid case study.

- **Machine Learning and Deep Learning Approach to Forecasting Household Electricity:** Developed ML & DL models for electricity demand forecasting, improving prediction accuracy and reducing computational time through feature engineering.
- **Bluetooth Based Home Automation:** Built Bluetooth-based home automation system using HC-05 module operating at 2.4 GHz, designed for elderly and physically challenged users to easily monitor and control household devices.
- **Design of A Multiband Minkowski Fractal Antenna for Wi-Fi, WiMAX:** Designed multiband Minkowski fractal antenna (3.3–7.8 GHz) with 4.2 dB peak gain, supporting Wi-Fi 802.11y (3.6 GHz), WiMAX (3.4–3.6 GHz), and WLAN 802.11 (5.31–6.32 GHz) applications.

Professional Experience

- **Walton Hi-Tech Industries PLC** **Dhaka, Bangladesh**
Senior Supply Chain Manager *2017 – 2022*
 - Led a progressive career path from *Assistant Engineer* to *Senior Manager*, overseeing cross-functional teams in supply chain, procurement, and product development.
 - Directed large-scale supply chain operations with focus on risk management, cost optimization, and strategic procurement, ensuring uninterrupted raw material availability during crises such as COVID-19.
 - Implemented data-driven demand forecasting models (ARIMA, ML-based approaches), reducing forecasting error variance by 15% and improving inventory turnover.
 - Negotiated supplier contracts and optimized procurement strategy, achieving up to 8-10% cost savings annually.
 - Led quality improvement initiative using Root Cause Analysis (RCA), reducing defect rates to below 0.2%, significantly enhancing customer satisfaction.
 - Supervised teams of up to 20 professionals, mentoring managers in project planning, KPI management, and new business development.

Publications

1. **K. B. Walid**, F. Ye, J. Ji, A. A. Ezzat, T. Miles, Y. L. Jiang, “Economic and Reliability Value of Improved Offshore Wind Forecasting in Bulk Power Grid Operation: A Case Study of The New York Power Grid,” *Applied Energy*, Submitted.
2. **K. B. Walid**, D. Gumina, L. Hobbs, Y. L. Jiang, Z. Shubo, K. Upadhyay, P. G. Kumar “A Risk Informed Capacity Analysis Commitment Tool for Reliable Bulk Power System Operation,” *2026 IEEE Power & Energy Society General Meeting (PESGM)*, Submitted.
3. **K. B. Walid**, Q. Li, Y. L. Jiang, “Value-Based Grid Modernization: Project Evaluation by Social Benefits,” *2025 57th North American Power Symposium (NAPS)*, In Press.
4. L. Hobbs, **K. B. Walid**, Y. L. Jiang, “Operational Resource Adequacy in Reliable Grid Operation with High Renewable Resources,” *2025 57th North American Power Symposium (NAPS)*, In Press.
5. **K. B. Walid**, R. Roach, Y. L. Jiang, K. Upadhyay, J. Meyer, P. G. Kumar, “Uncertainty Reserve Design of the New York Power Grid with High Offshore Wind Resources,” *2025 57th North American Power Symposium (NAPS)*, In Press.
6. **K. B. Walid**, Y. L. Jiang, “Short-Term Load Forecasting with Advanced Machine Learning Approaches: Spatial Variability and Model Performance,” *2024 56th North American Power Symposium (NAPS)*, El Paso, TX, USA, pp. 1–6, 2024. DOI: [10.1109/NAPS61145.2024.10741627](https://doi.org/10.1109/NAPS61145.2024.10741627).
7. D. Gumina, **K. B. Walid**, J. Meyer, P. G. Kumar, S. Zhang, “Reserve Procurement in the New York Power Grid with Offshore Wind Farms,” *2024 56th North American Power Symposium (NAPS)*, El Paso, TX, USA, pp. 1–6, 2024. DOI: [10.1109/NAPS61145.2024.10741830](https://doi.org/10.1109/NAPS61145.2024.10741830).

Certifications

- Programming for Everybody (Getting Started with Python) & Machine Learning by Coursera
- Productivity Improvement tools & technique and 5S Implementation-National Productivity Organization (NPO)

Membership and Awards

- Student Member, IEEE Power & Energy Society, IEEE Graduate Student Membership, IEEE Young Professionals
- Former Member, Electrical and Electronics Club, University of Dhaka
- 1st Runner-Up, Circuit Competition, Electronics & Communication Club, University of Dhaka
- Best Employee of the Month, Sourcing & Planning Division, Walton Hi-Tech Industries PLC

Appearances

- Volunteer: 2nd International Conference on Informatics, Electronics & vision(ICIEV), Dhaka, Bangladesh